

Advanced Pattern Classification Models for Intelligent Diagnosis of Jujube Leaf Diseases in Precision Agriculture

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Abstract—Jujube (*Ziziphus jujuba* Mill.), cultivated for more than seven thousand years, and its Indian counterpart ber (*Ziziphus mauritiana* Lamk.) are losing productivity to a group of foliar diseases led by *Alternaria* black spot, rust, anthracnose and phytoplasma-induced witches' broom. Learning-based diagnosis has matured for many field and horticultural crops, yet the jujube literature remains fragmented. This paper arranges the relevant work into five families: pathological characterisation, image-based recognition, severity quantification, attention-driven and multi-modal learning, and spray decision support. It also catalogues the image collections that jujube researchers can use today. Twenty representative studies are contrasted on crop, architecture, data, headline metric, grading ability and principal weakness, and a capability matrix records which of seven desirable properties each one offers. The analysis shows that no existing system combines jujube-specific validation, five-level severity scoring, automatic lesion-area measurement, text-assisted disambiguation and spray guidance, and that Indian cultivars are absent from the evidence base. Six open problems and a set of recurring technical challenges are formalised, and a roadmap is outlined around a regional benchmark, a single-pass hybrid CNN–Transformer, a symptom-lexicon vision–language extension, local pathogen confirmation and lightweight, decision-coupled deployment.

Index Terms—jujube, ber, *Ziziphus*, leaf disease recognition, severity grading, lesion segmentation, CNN–Transformer, vision–language model, precision agriculture.

I. INTRODUCTION

With a recorded cultivation history of over 7,000 years and a yearly global harvest above eight million tonnes, jujube (*Ziziphus jujuba* Mill.) ranks among the oldest orchard crops known, and a widening set of foliar and systemic pathogens now threatens it [1]. Leaf black spot causes the heaviest economic damage. Its principal agent is *Alternaria alternata* (Fr.) Keissler, and a second agent, the *A. arborescens* species complex, has lately been verified at a location beyond China for the first time [2]. Where infection is severe, early leaf drop and reduced photosynthetic capacity can remove 30–50% of annual yield [3]. Because conidia move on wind and splashing rain and need only two to three days to incubate, one growing season can host repeated rounds of reinfection [3]. Rust (*Phakopsora zizyphi-vulgaris*), anthracnose (*Colletotrichum gloeosporioides* and allied species [4]) and witches' broom (JWB) complete the list of major threats. JWB is associated with *Candidatus* Phytoplasma zizyphi, a bacterium confined to the phloem; it cannot be cured and kills a tree two to three years after infection [5].

In India the cultivated species is ber (*Ziziphus mauritiana* Lamk.), grown across Maharashtra, Rajasthan, Gujarat and Uttar Pradesh, where losses from these diseases fall largely on smallholders. The pathogens that operate in Indian orchards

have not been characterised, and no automated, image-based diagnostic tool has been reported for any ber cultivar. Growers instead depend on periodic walks by trained agronomists followed by fungicide sprayed at one uniform rate. That practice is subjective, demanding of labour and harmful to the environment, and it cannot respond to differences in disease intensity between orchard blocks or over the season [3], [6]. Fig. 1 shows typical black-spot symptoms on jujube foliage.



Fig. 1. Field photographs of jujube leaves bearing black-spot lesions, taken by the authors during orchard survey work.

Precision agriculture offers another model, in which cameras and learning algorithms track disease and drive variable-rate treatment. In jujube, however, uptake trails well behind rice, wheat, tomato and apple [6], [7]. The strongest jujube classifier published so far, a diffusion–transformer network, recognises six diseases with 93% accuracy, but it was built solely from Chinese forestry imagery and returns neither a severity grade, a lesion-area estimate nor a treatment suggestion [1]. An SSD-based detector for jujube disease spots works only on fruit photographed on an industrial sorting line [8], and a vision–language detector that attains 95.2% mAP on vegetable diseases has never been applied to jujube [9]. Molecular identification has been completed for European *Alternaria* isolates [2] but not for Indian strains, and nowhere has automated severity estimation been linked to recommended spray parameters for a jujube cultivar.

Against this background, the survey makes four contributions. It groups the relevant literature into a taxonomy that runs from pathology to treatment delivery; it tabulates twenty representative studies and the datasets usable for jujube research on shared criteria; it distils six research gaps and the technical challenges behind them; and it sets out concrete directions toward orchard-ready diagnosis. Existing approaches are classified in Section II and compared in Section

III. Gaps and challenges are formalised in Section IV, directions are proposed in Section V, and Section VI concludes.

II. LITERATURE REVIEW AND CLASSIFICATION OF EXISTING APPROACHES

Knowledge relevant to jujube leaf-disease diagnosis is spread across plant pathology, computer vision and agricultural engineering. To make it navigable, we sort it into the five families of Fig. 2 and add a further strand on the image collections currently available. Families A–E are ordered from the biology of the problem to the delivery of treatment.

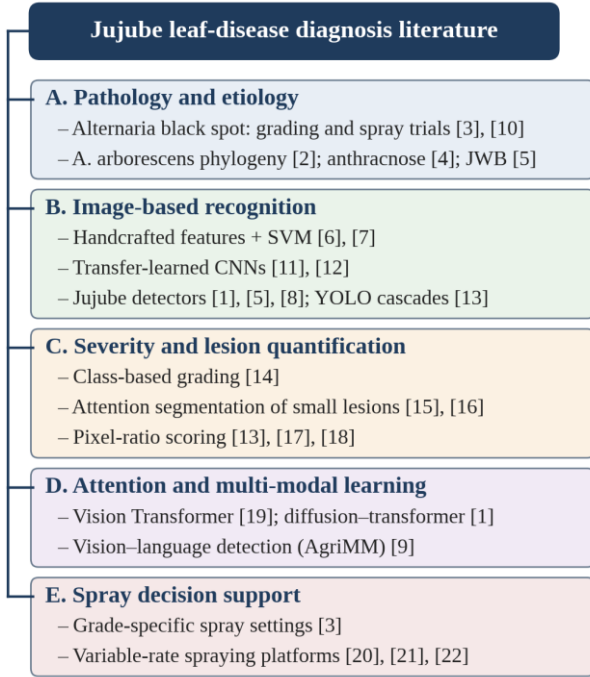


Fig. 2. Taxonomy of the literature reviewed in this survey.

A. Pathology and Etiology of Jujube Leaf Diseases

1) *Alternaria black spot*: Chinese studies regard black spot caused by *A. alternata* (Fr.) Keissler as the chief foliar disease of jujube. The first formal record on jujube foliage came from Aksu, Xinjiang, in 2012–2013, when epidemics stripped trees of leaves prematurely and reduced yield considerably [10]. The fungus enters through stomata; symptoms become visible after an incubation of two to three days, and conidia are released three to four days after a lesion forms. Wind and rain splash then carry inoculum to new tissue, so infection cycles follow one another, and incidence peaks once the rains begin [3].

The most detailed quantitative account comes from Zhang *et al.* [3]. Adopting the Xinjiang local standard DB65/T 4731-2023, they defined five grades by the diseased fraction of leaf area (Table I). They then marked 1,000 leaves on Dongzao trees in Hebei Province and compared nine spray configurations of 43% fluopyram + tebuconazole SC, formed by crossing three nozzle distances with three spray angles. No single configuration won everywhere: each grade had its own optimum, listed in Table I, and control efficiency at these optima lay between 97.15% and 99.52%. Activity of superoxide dismutase (SOD) and peroxidase (POD) rose together with control efficiency, which gives a physiological marker for judging whether a treatment has worked [3].

TABLE I
BLACK-SPOT SEVERITY GRADES AND GRADE-SPECIFIC OPTIMAL SPRAY SETTINGS

(after Zhang *et al.* [3], standard DB65/T 4731-2023)

Grade	Diseased leaf area	Optimal distance / angle
0	No lesion (prevention)	0.9 m / 0°
1	≤ 20%	1.2 m / 0°
2	20–25%	0.6 m / 40°
3	25–50%	0.6 m / 20°
4	> 50%	0.9 m / 40°

2) *Other major diseases*: Jujube rust, due to *P. zizyphi-vulgaris*, raises orange-yellow uredia beneath the leaf, on its abaxial side, and heavy infection may strip a tree of foliage. Anthracnose, due to *C. gloeosporioides*, leaves sunken circular lesions on both leaves and fruit; a Thai study of Indian jujube (*Z. mauritiana*) recently reported, for the first time, that *C. fructicola* and *C. siamense* can cause the disease [4]. JWB, attributed to *Ca. Phytoplasma zizyphi*, is the most destructive of the group. It triggers shoot proliferation and leafy flowers, and a tree succumbs two to three years after infection. Unlike the fungal diseases, it is spread by leafhoppers and through grafting, which makes early visual recognition especially valuable [5].

3) *Visual confusability*: In orchards these diseases often coexist and resemble one another. Black spot and rust can both appear as small dark spots, anthracnose lesions can mimic advanced black spot, and the earliest JWB sign, shoot proliferation, is expressed on the whole tree rather than on single leaves. Such overlap argues for recognisers that draw on context as well as pixels, an idea proven for vegetables by the AgriMM framework [9] but not yet for any fruit tree.

B. Image-Based Recognition

1) *Handcrafted features*: First-generation systems computed engineered descriptors, namely colour histograms, GLCM and LBP texture statistics and shape measures, and passed them to classifiers such as the support vector machine (SVM). Accuracy that was acceptable under laboratory control did not survive the move to orchards, where illumination changes, leaves overlap and occlude one another, and backgrounds are cluttered [6], [7].

2) *Transfer-learned CNNs*: After AlexNet’s 2012 ImageNet result, deep convolutional networks took over the field [6], [7]. The usual recipe now fine-tunes an ImageNet-trained backbone such as ResNet, DenseNet, EfficientNet, Inception or MobileNet on crop images, so that high accuracy is reachable from modest data. Goyal and Lakhwani [11] benchmarked EfficientNetB0, ResNet50, DenseNet121 and InceptionV3 on 759 citrus images in nine classes, among them black spot, canker and greening, and DenseNet121 led at 96.4%. Jannat *et al.* [12] reached 96.8% on jute leaf diseases with an explainable lightweight CNN refined by semi-supervised self-training. Because that scheme lowers the demand for labelled examples, it suits crops whose annotated imagery is scarce.

3) *Jujube-specific detectors*: Only three learned image models address jujube directly. Yin *et al.* [8] used the Single Shot MultiBox Detector (SSD) to find defects on jujube fruit passing through an automated sorter and obtained 89.3% precision. The work proves that real-time deep detection is practical on an industrial line, but leaves lie outside its scope.

Yan *et al.* [1] went further with a diffusion–transformer model equipped with parallel attention for joint growth estimation and disease detection in jujube forests. Trained on field images gathered in Bayannur, Inner Mongolia, between March and July 2023, it reached 93% accuracy and 95% precision when separating six classes (rust, powdery mildew, anthracnose, black spot, date mania and seedling stem rot), surpassing baselines built on AlexNet, ResNet, SVM and Random Forest. Its size of roughly 95 M parameters, its lack of severity output and its restriction to Chinese forest conditions nonetheless hinder transfer to Indian ber orchards. For JWB, Guo *et al.* [5] built a cascade that joins a region proposal network (RPN), Chan–Vese segmentation and transfer learning, and reported 83.57% accuracy on images with complex backgrounds. The comparatively low figure reflects how hard canopy-level symptoms are to read in real orchards and calls for models with stronger contextual reasoning.

4) *YOLO detectors and cascades*: Single-stage detectors of the YOLO (You Only Look Once) line are favoured for localising symptoms because they infer in real time. By pairing multi-scale feature fusion with attention, YOLOv8 has become the reference detector in agricultural work and exceeds 90% mAP on several crop–disease datasets [9], [13]. Patel *et al.* [13] assembled a four-model cascade for drone imagery: YOLOv8 proposes regions of interest, DeepLabV3+ removes the background, a CNN names the disease (96.97% accuracy) and U-Net marks damaged pixels (99%) for severity evaluation. The numbers are strong, but executing four networks in sequence is too costly for real-time inference on the low-cost edge hardware found in Indian orchards.

5) *Review-level evidence*: Two broad reviews frame these results. Ahmad *et al.* [6] examined 160 articles published from 2020 to 2024 on CNN-based classification, detection and segmentation and found a mean accuracy of 95.37%. A 2025 review in *Frontiers in Plant Science* [7] covered one- and two-stage detectors and attention mechanisms, with reported accuracies reaching 98.31%. Neither review includes experimental work on jujube.

C. Severity Grading and Lesion Quantification

Severity grading places disease intensity, usually the share of tissue affected, on a discrete or continuous scale. A grade says more than a healthy/diseased verdict: it indicates whether prevention, cure or intensive intervention is appropriate, and it lets dose and application settings be tuned. Zhang *et al.* [3] made the case concretely, since their grade-dependent optima show that one fixed protocol is suboptimal and may even harm plant physiology. If A_L is the lesion area and A_T the total leaf area visible in an image, the severity percentage is

$$S = (A_L / A_T) \times 100\% \quad (1)$$

and the grade follows by thresholding S at the boundaries of Table I. We classify existing methods into three groups according to how they arrive at S or at the grade.

1) *Class-based grading*: The simplest route treats every grade as a class label. Liu *et al.* [14] applied a two-stage CNN to apple *Alternaria* leaf blotch and assigned four severity stages at 94.8% accuracy. Because the pathogen genus is shared with jujube black spot, the design is directly relevant, although it resolves four stages rather than five and only approximates the area fraction behind each label.

2) *Attention-enhanced segmentation of small lesions*: Early Grade 1 black spot consists of tiny scattered lesions that plain encoders tend to miss. Xu *et al.* [15] inserted the CBAM attention block into a DFL-UNet trained with a hybrid loss. On apple *Alternaria* blotch and brown spot the model reached 91.07% MIOU and 95.16% F1, a gain of 1.96 points in MIOU over a standard U-Net. Liu *et al.* [16] obtained 88.7% MIOU for small apple-leaf spots with an improved MA-Net built around multi-scale attention. Taken together, these results suggest that attention modules of this kind fit the dispersed lesions of low-grade jujube infection.

3) *Pixel-ratio segmentation and field scoring*: Networks that evaluate (1) directly from pixel masks give the finest estimate, because they avoid the coarse binning implicit in class labels. Wang *et al.* [17] coupled an improved HRNet with a DRL-watershed step for apple leaf severity grading, reached 94.2% segmentation accuracy across four diseases on 417 images, and showed that separating touching lesions matters. Larsen *et al.* [18] delivered an end-to-end scorer for early and late peanut leaf spot that works on plot-level smartphone photographs (F1 = 0.91), replacing subjective expert ratings with algorithmic ones. Running on ordinary phones makes it attractive for resource-limited farmers, although it cannot tell disease types apart. The U-Net stage of Patel *et al.* [13] likewise supplies pixel-level damage for severity evaluation.

Despite this progress, deep segmentation has never been used on jujube black spot to evaluate the $\leq 20\%$, 20–25%, 25–50% and $> 50\%$ boundaries of DB65/T 4731-2023. Zhang *et al.* [3] relied on semi-automatic LabelMe annotation, which becomes impractical at the scale of a field survey.

D. Attention-Driven and Multi-Modal Learning

1) *Vision Transformers*: Since their introduction in 2017, Transformers have increasingly supplemented, and for some tasks replaced, purely convolutional designs in vision, plant disease analysis included. Self-attention relates distant regions of an image and can therefore describe how disease spreads over a whole leaf, which complements the sensitivity of convolutions to local edges and texture. Mwaibale *et al.* [19] trained ViTs on Fast Gradient Sign Method (FGSM) adversarial examples to recognise rust and anthracnose in common bean. They reached F1 = 0.93 and found the models more robust to background variation than CNN baselines. In the jujube study of Yan *et al.* [1], the diffusion component generated augmented samples while transformer blocks modelled global features.

2) *Vision–language detection*: Coupling images with language is the most consequential recent development. Liu and Wang [9] designed AgriMM, an agricultural multi-modal detector trained on 30,000 images spanning 30 disease categories of five vegetables. It scored 95.2% mAP, an improvement of 11.6% over the strongest image-only baseline. Three modules underpin it. HCACB is a backbone whose convolutional and attention branches cooperate to capture local texture and global lesion distribution in parallel; CVL-PAN is a context-enhanced path-aggregation network that merges visual and linguistic features at several scales; and AR-TCL is an adaptive contrastive objective that aligns image regions with their text. Grad-CAM visualisations revealed that the descriptions act as semantic guides, drawing attention to diagnostic lesions and reducing confusion between similar-looking diseases. This is exactly the difficulty that separates

black spot from rust and anthracnose on jujube. At 48.7 frames per second, and with 80.0% mAP when transferred to public datasets with no fine-tuning, AgriMM also generalises well across domains, yet it has only ever been trained and tested on vegetables [9].

E. Spray Decision Support

A diagnosis creates value only when it changes treatment. Chen *et al.* [20] designed a variable-spray UAV for rice pest and disease areas in which a back-propagation (BP) neural network regulates flow; wind-tunnel and field tests kept flow error below 20%, but the system does not analyse disease imagery. Lu *et al.* [21] fused LiDAR, RGB imaging and machine learning to set variable spray rates in wheat and estimated biomass with less than 15% error, so its input is canopy biomass rather than leaf-level disease grade. Farooque *et al.* [22] field-tested a deep-learning smart variable-rate sprayer and reported $F1 = 0.955$, with a focus on pests and without severity grading or Indian crop data. For jujube, Zhang *et al.* [3] supply the missing agronomic link in the form of grade-specific spray settings, yet those settings have never been driven by an automatic detector.

F. Data Resources

Table II lists the image collections that can support jujube modelling today, together with the role each could play. None offers graded leaf images of Indian ber. The largest general resource, PlantVillage [23], is suited to pre-training, while the jujube-specific sets are either small, restricted to fruit, or unreleased [1], [3], [24], [25], [26].

TABLE II
IMAGE DATASETS RELEVANT TO JUJUBE DISEASE RESEARCH

Dataset	Size and classes	Potential role
Tropical fruit leaf disease (jujube, star fruit, guava) [24]	1,602 original / 7,885 augmented; defect leaf, healthy, insect feeding, leaf curl	Baseline comparison; transfer-learning pre-training
JujubeDataSet / JujubeSSD [25]	Size not disclosed; black spot on fruit surface	Cross-domain testing; encoder pre-training
Jujube forest disease set [1]	Not public; rust, powdery mildew, anthracnose, black spot, date mania, stem rot	Architecture benchmark (diffusion-transformer)
PlantVillage [23]	54,306 images; 38 classes, 14 crops, 26 diseases	Backbone pre-training; generalisation baseline
JujubeBruiseNet (Z. mauritiana) [26]	1,464 images; healthy vs. bruised fruit	Familiarity with ber visual traits
Zhang <i>et al.</i> leaf images [3]	1,000 leaves; black spot Grades 0–4	Grading reference; physiological labels

III. COMPARATIVE ANALYSIS

A. Multi-Criteria Comparison

Table III sets twenty studies side by side: jujube pathology, general deep-learning diagnosis, severity grading, precision spraying and two reviews. For every entry it records the method, the crop and disease targeted, the data used, the headline result, any severity grading and the chief limitation. Table IV condenses the same evidence into a capability matrix over seven properties that an orchard-ready jujube system would need.

TABLE III
MULTI-CRITERIA SUMMARY OF SURVEYED STUDIES ON JUJUBE AND ALLIED CROPS

Study	Crop / disease	Method / architecture	Data	Reported result	Grading	Principal limitation
Yan <i>et al.</i> , 2024 [1]	Jujube / 6 diseases	Diffusion-transformer with parallel attention	Field images, Inner Mongolia	Acc. 93%; precision 95%	No	About 95 M parameters; no grading; Chinese forest data
Papp <i>et al.</i> , 2026 [2]	Jujube / leaf spot (<i>A. arborescens</i>)	Multilocus phylogeny (ITS, TEF1- α , Rpb2)	3 isolates, 2 cultivars	Pathogenicity confirmed	No	Laboratory work without imaging; European isolates only
Zhang <i>et al.</i> , 2026 [3]	Jujube / Alternaria black spot	Grade-wise spray-parameter trials; SOD, POD, MDA assays	1,000 leaves, 5 grades	Control efficiency 97.15–99.52%	Yes (0–4)	Semi-automatic lesion labelling; no learning model; Chinese orchards only
Guo <i>et al.</i> , 2023 [5]	Jujube / witches' broom	RPN + Chan-Vese + transfer learning	Diseased-tree images	Acc. 83.57%	No	Modest accuracy; tree level only; multi-stage design
Ahmad <i>et al.</i> , 2024 [6]	Many crops / 160 studies	Systematic review: CNN classification, detection, segmentation	160 articles (2020–2024)	Mean acc. 95.37%	Partial	Review without new model; jujube not covered
Frontiers Review, 2025 [7]	Many crops / leaf diseases	Review: one- and two-stage detectors, attention	Meta-analysis	Up to 98.31%	No	Review only; no experimental validation
Yin <i>et al.</i> , 2024 [8]	Jujube fruit / disease spots	SSD detector	Sorting-line images	Precision 89.3%	No	Fruit on a sorter; leaves not addressed
Liu & Wang, 2026 [9]	5 vegetables / 30 diseases	AgriMM (HCACB, CVL-PAN, AR-TCL), image + text	30,000 images	mAP 95.2%	No	Vegetables only; no grading; high compute demand
Goyal & Lakhwani, 2025 [11]	Citrus / black spot, canker, greening	EfficientNetB0, ResNet50, DenseNet121, InceptionV3	759 images, 9 classes	DenseNet121: 96.4%	No	Small single-crop set; no segmentation
Jannat <i>et al.</i> , 2025 [12]	Jute / leaf diseases	Lightweight CNN + semi-supervised self-training	Limited labelled data	Acc. 96.8%	No	Tropical fibre crop; no grading
Patel <i>et al.</i> , 2024 [13]	Vegetables / several diseases	YOLOv8 $\rightarrow$ DeepLabV3+ $\rightarrow$ CNN $\rightarrow$ U-Net	Drone images	Cls. 96.97%; seg. 99%	Yes	Four chained models; drone-dependent; no spray coupling
Liu <i>et al.</i> , 2022 [14]	Apple / Alternaria leaf blotch	Two-stage CNN	Apple disease images	Acc. 94.8% (4 stages)	Yes (4)	Four stages instead of five; apple only
Xu <i>et al.</i> , 2023 [15]	Apple / Alternaria, brown spot	DFL-UNet + CBAM, hybrid loss	Apple leaf images	MIoU 91.07%; F1 95.16%	No	Apple only; neither grades nor treatment link
Liu <i>et al.</i> , 2024 [16]	Apple / Alternaria blotch, brown spot	Improved MA-Net, multi-scale attention	Field images	MIoU 88.7%	No	Apple only; masks not turned into severity
Wang <i>et al.</i> , 2025 [17]	Apple / 4 leaf diseases	Improved HRNet + DRL-watershed	417 images	Seg. acc. 94.2%	Yes	Limited data; apple only; no spray guidance

Study	Crop / disease	Method / architecture	Data	Reported result	Grading	Principal limitation
Larsen <i>et al.</i> , 2025 [18]	Peanut / early and late leaf spot	End-to-end segmentation pipeline	Plot-level phone images	F1 = 0.91	Yes (score)	Peanut only; no disease-type output or treatment advice
Mwaibale <i>et al.</i> , 2025 [19]	Common bean / rust, anthracnose	ViT + FGSM adversarial training	Field images, Tanzania	F1 = 0.93	No	Two classes; regional data; no grading
Chen <i>et al.</i> , 2019 [20]	Rice / pest and disease areas	BP neural network for variable-spray UAV	Wind-tunnel and field data	Flow error < 20%	No	No image-based diagnosis; spraying only
Lu <i>et al.</i> , 2025 [21]	Wheat / variable-rate spraying	LiDAR + RGB + machine learning	Farmland imagery	Biomass error < 15%	No	Biomass input; no leaf-level disease grading
Farooque <i>et al.</i> , 2023 [22]	Multiple crops / pests	Deep-learning smart variable-rate sprayer	Field-trial data	F1 = 0.955	No	Pest-oriented; no severity; no Indian crop data

B. Capability Matrix

The columns of Table IV are: JD, validated on jujube data; DL, a learned image model; MC, multi-class disease identification; SG, severity grading; LS, automatic lesion segmentation; MM, use of text or context alongside images; and SP, spray guidance. The two reviews [6], [7] are omitted because they report no system of their own.

TABLE IV
CAPABILITY MATRIX OF THE SURVEYED SYSTEMS

Study	JD	DL	MC	SG	LS	MM	SP
[1]	●	●	●	–	–	–	–
[2]	●	–	–	–	–	–	–
[3]	●	–	–	●	○	–	●
[5]	●	●	–	–	–	–	–
[8]	○	●	–	–	–	–	–
[9]	–	●	●	–	–	●	–
[11]	–	●	●	–	–	–	–
[12]	–	●	●	–	–	–	–
[13]	–	●	●	●	●	–	–
[14]	–	●	–	●	–	–	–
[15]	–	●	●	–	●	–	–
[16]	–	●	●	–	●	–	–
[17]	–	●	●	●	●	–	–
[18]	–	●	–	●	●	–	–
[19]	–	●	●	–	–	–	–
[20]	–	○	–	–	–	–	●
[21]	–	○	–	–	–	–	●
[22]	–	●	–	–	–	–	●

● present; ○ partial (fruit-only data, non-image learning or semi-automatic labelling); – absent.

C. Evaluation Metrics

The headline figures in Table III use different metrics, and these must be understood before any comparison. With TP, TN, FP and FN denoting true positives, true negatives, false positives and false negatives,

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (2)$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (3)$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (4)$$

$$\text{F1} = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \quad (5)$$

Accuracy summarises overall correctness but can mislead when classes are imbalanced, as severity grades usually are. Precision tells how many detections are genuine, recall how many diseased samples are found, and F1 balances the two; the area under the ROC curve (AUC-ROC) expresses how well classes

separate across thresholds. Detectors are reported by mAP [1], [9], computed at an IoU threshold of 0.5 (mAP@0.5) or averaged over 0.5:0.95, whereas segmentation models are reported by MIoU [15], [16] or pixel accuracy [13], [17]. Throughput in frames per second (FPS) indicates suitability for real-time use [9].

D. Observations

1) *Crop coverage*: only five of the twenty studies concern jujube [1], [2], [3], [5], [8], just three of these apply learned image models [1], [5], [8], and none uses Indian material.

2) *Severity*: five studies grade severity [3], [13], [14], [17], [18], but only one follows the five-level jujube standard [3], and it does so without a learning model.

3) *Modality*: AgriMM [9] is the sole multi-modal system; every other entry relies on pixels alone.

4) *Treatment link*: the spraying studies [20], [21], [22] tune application but ignore disease grade, while the one study that relates grade to spray setting [3] has no detector.

5) *Deployability*: resource demands range from phone-ready scoring [18] and 48.7 FPS detection [9] to a classifier of about 95 M parameters [1] and a four-network cascade [13].

6) *Comparability*: reported values span 83.57% [5] to 99% [13], yet they measure different quantities on different data, so no ranking is meaningful until a shared benchmark exists.

No row in Table IV is filled across all seven columns, and no jujube system reaches beyond three of them.

IV. RESEARCH GAPS AND CHALLENGES

A. Open Research Gaps

G1: Regional evidence. Black spot has scarcely been examined in Indian agroecosystems. Published characterisations come from China [3] and Europe [2], so the pathogen diversity arising under Indian climates and cultivation practices is unknown.

G2: Standardised severity output. Deep models for jujube stop at disease presence or coarse categories [1], [5], [8]. Five-level grading tied to an agricultural standard such as DB65/T 4731-2023 [3] has not been attempted.

G3: Automatic lesion measurement. Lesion area for jujube black spot is still obtained through manual or semi-automatic annotation [3], which cannot scale to orchard-wide or season-long monitoring.

G4: Diagnosis-to-treatment coupling. Grade-specific spray settings have been established experimentally [3] and variable-rate hardware exists for other crops [20], [21], [22], but no AI-driven recommender turns a detected grade into an application setting.

G5: Multi-modal diagnosis. Text- and context-aware detection has been validated only on vegetables [9]. For jujube, where co-occurring diseases look alike, no study uses complementary non-visual information to raise accuracy or interpretability.

G6: Open benchmark data. No public leaf-disease dataset represents Indian ber cultivars, field conditions and pathogen variability; the collections in Table II cover other regions, fruit surfaces or unrelated symptom classes.

B. Recurring Technical Challenges

Several obstacles recur across the reviewed studies and will shape any solution. First, inter-class similarity and co-infection blur the boundaries between black spot, rust and anthracnose, which image-only models find hard to resolve [9]. Second, field imagery varies in illumination, occlusion, leaf overlap and background clutter [6], [7], so robustness to background change becomes a design goal [19]. Third, early lesions are small and sparse, demanding attention or multi-scale mechanisms [15], [16]. Fourth, pixel-level annotation is expensive, which motivates semi-supervised learning [12] and better tooling than semi-manual labelling [3]. Fifth, compute budgets on farm devices are tight compared with large or cascaded models [1], [13], although phone-level deployment has been shown for peanut [18]. Finally, heterogeneous metrics and datasets, combined with class imbalance across grades, make fair comparison difficult.

V. FUTURE RESEARCH DIRECTIONS

A. A Regional, Richly Annotated Benchmark

The first priority is a public ber/jujube leaf dataset gathered from commercial orchards and smallholder plots in Maharashtra. The region is semi-arid, with 400–900 mm of rain concentrated in the monsoon, and its humid post-rain spells favour *Alternaria* sporulation and reinfection [3]. Sampling should span growth stages (pre-flowering, fruiting, post-harvest), the June–September peak of infection and the low-incidence dry months of November–February. Healthy Grade 0 leaves from the same orchards belong in the set as a reference class that curbs false positives. Both leaf surfaces merit imaging, since early sporulation shows on the abaxial side. Capturing images with consumer smartphones and entry-level DSLRs at 15–40 cm under natural light would match the conditions of eventual users. Each record could hold images of at least 1024×1024 pixels, LabelMe polygons for leaf and lesions, YOLO-format boxes, the lesion percentage from (1), a pathologist-validated grade, a co-infection flag, and metadata on GPS location, date, season, cultivar, tree age and weather. Roughly 300–400 original images collected for each grade could be expanded through flipping, $\pm 30^\circ$ rotation, $\pm 20\%$ brightness and contrast change, Gaussian noise, CutOut, CutMix and Mosaic, and then split 80:20 for training and testing. Such a resource would do for ber what PlantVillage [23] has done for other crops.

B. Controlled Benchmarking of Existing Architectures

With shared data in place, representative CNNs (ResNet, DenseNet, EfficientNet), detectors (YOLOv5, YOLOv8,

YOLOv11), transformers (ViT, Swin Transformer) and hybrids should be compared on identical splits, starting from ImageNet weights as in [11]. Reports should give accuracy together with FPS, parameter count and mobile inference time, so that deployment feasibility is judged alongside accuracy.

C. Single-Pass Hybrid CNN–Transformer Models

A convolutional branch, for instance EfficientNet or ResNet, can capture local lesion texture while multi-head self-attention models how disease spreads across the leaf, with an adaptive weighting module balancing the two streams [7], [9]. Adaxial and abaxial images could enter as paired inputs, and separate heads could emit the disease class, the five-level grade and a lesion mask from which (1) is computed [14], [17], [18]. Joint training might minimise

$$L = L_{CE} + L_{MSE} + L_{Dice} \quad (6)$$

where the terms are cross-entropy for classification, mean-squared error for severity regression and Dice loss for segmentation.

D. Vision–Language Extension with a Symptom Lexicon

To separate look-alike diseases, the visual model can be aligned with expert text through region–text contrastive learning of the AR-TCL type, which gives image features a stable semantic anchor [9]. A two-layer knowledge base is suggested. Its first layer is a symptom lexicon holding discriminative descriptors of lesion morphology, colour, boundary type and conidial visibility for every disease–grade pair; its second records environmental context such as season of infection, humidity triggers and co-infection indicators. A pathologist panel should validate every entry. A pre-trained CLIP ViT-B/32 text encoder could first stay frozen while the visual backbone adapts and later be fine-tuned on the lexicon, and both early and late fusion deserve testing. A weighted objective

$$L = L_{det} + 0.5 L_{con} + 0.5 L_{seg} \quad (7)$$

would balance detection, contrastive and segmentation terms. Ablations comparing visual-only, text-only and fused inputs, measured by mAP@0.5, mAP@0.5:0.95, precision, recall, F1 and FPS, would then quantify what language contributes.

E. Confirming Local Pathogens

Image labels are only as reliable as the etiology behind them. Multilocus phylogenetics based on ITS, TEF1- α and Rpb2, already carried out for European isolates [2], should be extended to *Alternaria* from Indian ber so that dataset classes rest on confirmed causal agents.

F. Decision-Coupled, Edge-Ready Deployment

The grade-to-setting relations in Table I [3] can act as a first lookup that converts a predicted grade into nozzle distance and angle, and existing variable-rate platforms [20], [21], [22] show how such commands could be carried out. Models kept light enough for smartphones, as demonstrated for peanut [18], and fast enough for real-time inference [9] would bring the loop from diagnosis to treatment within reach of small orchards. Fig. 3 shows how directions A, C and D can be chained into one pipeline.

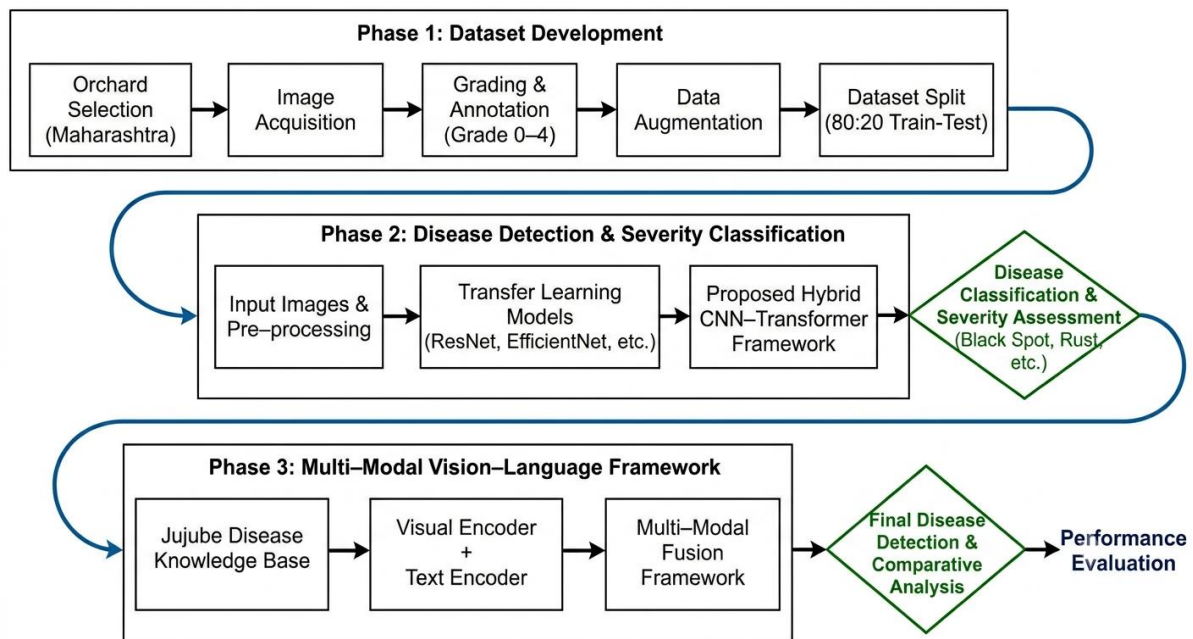


Fig. 3. Conceptual three-phase pipeline that links dataset development, hybrid CNN–Transformer classification and vision–language fusion.

VI. CONCLUSION

This survey has reviewed jujube leaf-disease diagnosis against the wider progress of learning-based plant pathology. Work on cereals, vegetables, apple and citrus has advanced from handcrafted features through transfer-learned CNNs and real-time detectors to attention-based segmentation, transformers and vision–language fusion, with measurable gains at each step. Jujube has shared in few of these advances. Its image-based studies address fruit on a sorting line, one canopy-level disease or forest imagery from China, and none reports a standard severity grade. Our comparison of twenty studies found no system that validates on jujube, grades severity on five levels, measures lesion area automatically, exploits text or context and recommends spray settings, and none that draws on Indian ber data. A public Maharashtra benchmark, single-pass hybrid CNN–Transformer models, symptom–lexicon vision–language alignment, molecular confirmation of local pathogens and lightweight, decision-linked deployment therefore stand out as the most productive next steps toward precision management of jujube orchards.

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